Unsupervised Anomaly Detection in Mars HiRISE Orbital Imagery

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OrbitIQ

Final answers

- **Model:** v3 - convolutional autoencoder trained from scratch; linear bottleneck → fixed-length 256-d 1-D latent; loss = 0.84·(1−SSIM) + 0.16·L1 + 0.1·Gradient-L1; map-projection no-data wedges masked
- **Novelty Score:** Isolation Forest s(x,n) on latents (higher = more anomalous), 500 trees × 5 seeds
- **Statistical threshold:** modified z-score > 3.5 (median + 3.5·MAD/0.6745) = **0.5293**
- **Crops flagged: 86** of 10422 (0.83 %)
- **Top-5 analysed in Phase 3:** sample_00849.jpg (0.587), sample_06917.jpg (0.582), sample_05849.jpg (0.580), sample_03423.jpg (0.579), sample_03383.jpg (0.577)

Key findings

1. The decisive fix was **not** the loss but the data. Masking the map-projection no-data wedges (v3) removed a nuisance mode that had made every v1/v2 flag a border crop. It turned a bimodal score distribution into a unimodal one with a genuine tail.
2. The flagged set contains **clear non-Martian content** (an aerial photograph of the Pentagon, perfectly periodic grids), recognisable in Phase 3 by diffuse whole-frame reconstruction error. It also contains **rare genuine terrain** (north-polar layered deposits, fresh deep craters), which the metadata join (2.3) identifies as a terrain class concentrated in two polar observations.
3. The spatial bottleneck (v4) was tried and **rejected**, because its coordinates encoded local brightness.
4. **Limitations:** small local sensor defects are missed by latent-space scoring (the residual catches two of three), and recall for terrestrial content is incomplete (sample_05508).

0. Data and sanity pass

Sanity-pass findings

- 10,422 single-channel 227×227 JPEG crops from 172 source observations; every crop maps to exactly one `source_image_id` and every id has metadata (checked with `assert`).
- Pixel values span the full 0-255 range; we rescale by 1/255 to [0, 1] and use a sigmoid decoder output.
- **≈20 % of crops contain a solid-black no-data wedge** (the map-projected edge of the HiRISE strip). These are *normal* data, but they are the highest-contrast edges in the whole dataset, which matters for both the reconstruction loss and the anomaly detector (see changelog).
- Very bright crops (frost / dust) exist and are genuine terrain, so a brightness heuristic cannot separate anomalies - consistent with the organisers' warning.

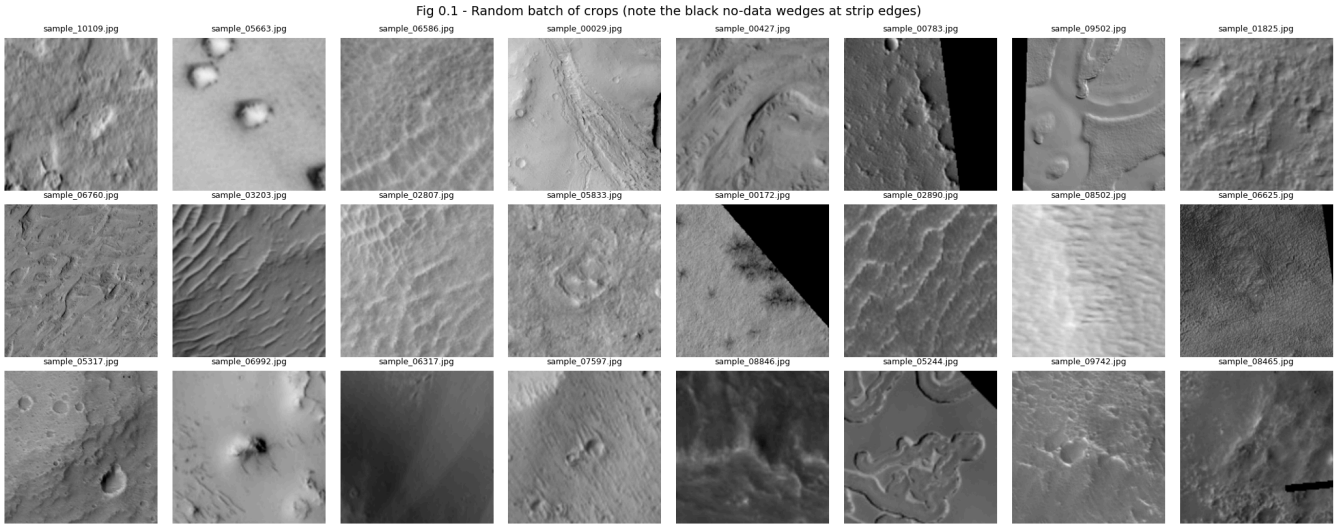


Fig 0.1 - Random batch

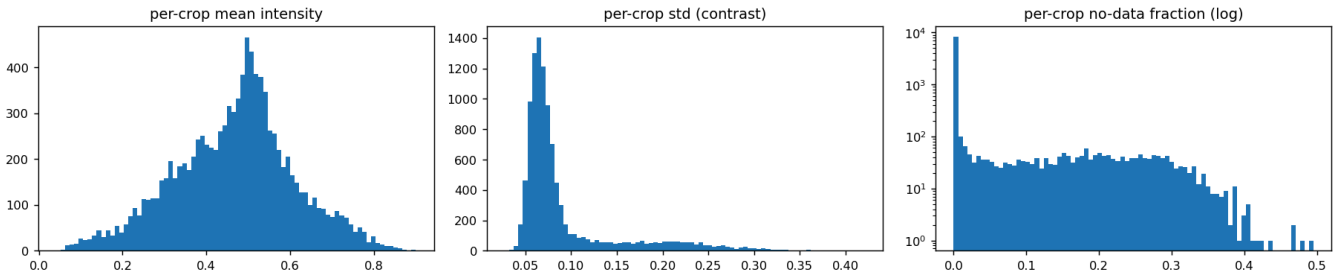


Fig 0.2 - Per-crop intensity, contrast and no-data statistics

0.2 No-data (map-projection) masking - introduced in v3

v1 and v2 showed that the black map-projection wedges, which are *normal* data, dominated both the reconstruction loss and the latent space (see changelog). From v3 on, each crop gets a **no-data mask**. A pixel is masked if it belongs to a near-black (≤ 5) region that

- survives a 5×5 morphological opening, so thin dark lines are *not* masked. A dead sensor column or a black streak could itself be an injected anomaly, and hiding it would be wrong.
- touches the crop border, because strip edges enter from outside the crop.
- covers at least 0.5 % of the crop.

The mask is dilated by 2 px to cover JPEG ringing along the wedge edge. Masked pixels are filled with the median of the crop's valid pixels, which gives a flat, texture-free region with no razor-sharp edge. They are also **excluded from the loss and from every reconstruction-error metric and heatmap**. The filled image, not the original, is what the v3 encoder sees.

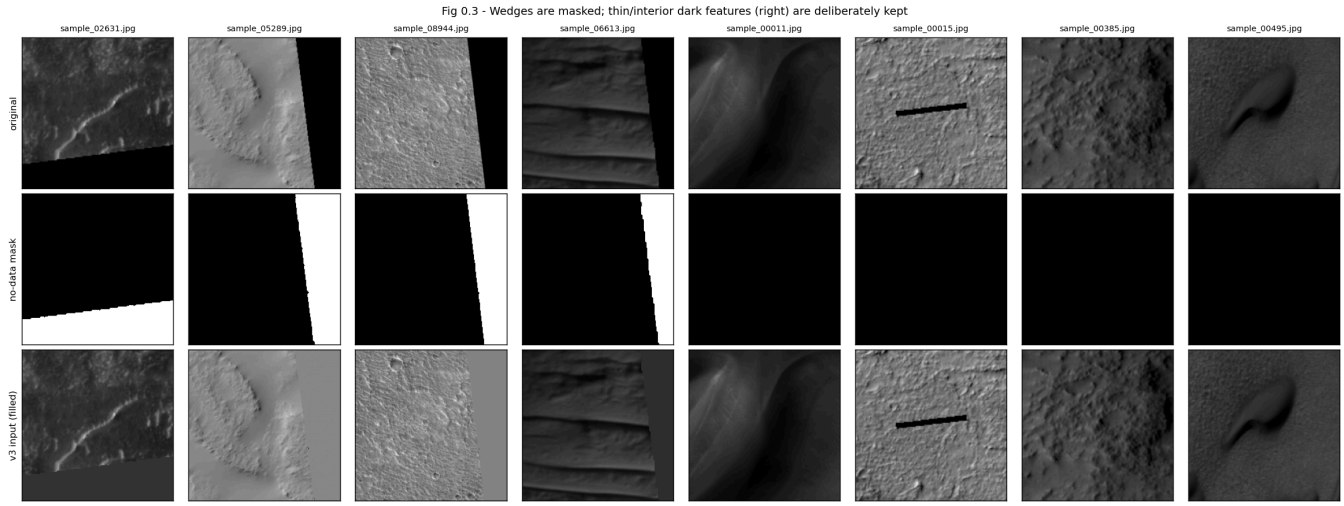


Fig 0.3 - No-data masks: projection wedges are masked; thin interior dark features are deliberately kept

Phase 1 - Deep Latent Compression

1.1 Architecture design

Design. A convolutional autoencoder written from scratch in PyTorch (no pretrained weights, no transfer learning).

- **Encoder:** five stride-2 3×3 convolutions (32→64→128→256→256 channels), BatchNorm + LeakyReLU. 227 → 114 → 57 → 29 → 15 → 8. Five halvings are the minimum that brings the map to 8×8, so the deepest units have a receptive field covering a large part of the crop (whole craters, dune trains) before the bottleneck.
- **Bottleneck (final model, v3):** flatten (256×8×8 = 16,384) → `Linear` → **fixed-length 1-D latent vector of 256 floats** (the Phase 2 input). The global linear layer lets every latent coordinate combine evidence from the whole crop, so the code describes *what kind of scene* this is (terrain type, texture class, relief).
- **Alternative tried and rejected (v4):** a *spatial* bottleneck, a 1×1 convolution to an 8×8 grid × 4 channels flattened to the same 256-d vector. The aim was to keep small local defects visible, but each coordinate then mostly encoded *local brightness*, so the Isolation Forest isolated very bright or dark scenes instead of anomalies (see changelog). We therefore kept the global code.
- **Decoder:** mirror image: `Linear` (v4: 1×1 conv) back to the 8×8×256 map → five ×2 transposed-conv stages to 256×256, centre-crop to 227×227, sigmoid. Cropping (rather than odd output-padding) keeps every stage a clean ×2.
- Stride-2 convolutions replace pooling so the network learns its own downsampling.
- The bottleneck type, latent size, loss and no-data handling are configurable because they change across v1 → v4 (`CONFIGS`).

The latent size trades **reconstruction fidelity** (a larger code keeps more detail) against **outlier separability**. An Isolation Forest splits on randomly chosen coordinates. If many latent dimensions carry redundant or near-constant information, most random splits are wasted and the isolation depth of true outliers gets diluted. We trained the final configuration with d ∈ {64, 128, 256, 512} (short schedule) and measured three things:

- held-out SSIM (fidelity)
- the number of principal components needed for 90 % of latent variance (how much of the code is used)
- a label-free separability proxy: the *tail ratio* (99.9th-percentile Novelty Score − median) / MAD. This measures how far the most isolated crops sit from the bulk, in robust units.

latent_dim	epochs	val_mse	val_ssim	pca90	tail_ratio	n_flag_mad35
64	12	0.0027	0.6822	30	10.8869	868
128	12	0.0029	0.6781	8	9.5130	641
256	12	0.0023	0.6920	85	11.0490	828
512	12	0.0023	0.6961	109	10.4945	687

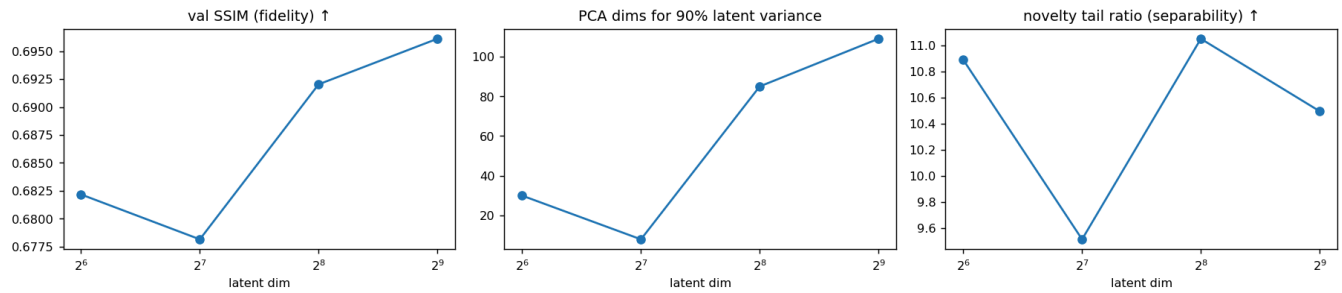


Fig 1.1 - Latent dimensionality sweep

Choice: d = 256. The sweep uses a short 12-epoch schedule and a single seed, so its flagged counts are not comparable with the fully trained model and are shown only for completeness. The differences are indicative, not conclusive.

- **Fidelity saturates.** Held-out SSIM is 0.682 (64), 0.678 (128), 0.692 (256) and 0.696 (512). Doubling from 256 to 512 buys only +0.004.
- **Effective dimensionality grows sub-linearly.** 30, 8, 85 and 109 principal components hold 90 % of the latent variance. Going to 512 adds 256 coordinates but only about 24 effective directions. The rest are redundant, and redundant coordinates **dilute an Isolation Forest**, whose splits pick a coordinate uniformly at random.
- **Separability peaks at 256.** The tail ratio is 10.9, 9.5, 11.0 and 10.5. The correlation with the no-data fraction (the nuisance removed in v3) creeps up with size (0.03 → 0.08), because a bigger code has room to re-encode fill geometry.
- **d = 128 is an outlier.** 90 % of its variance sits in only 8 components, a partial collapse under the short schedule, and it has the weakest tail.

So 256 is the knee of the trade-off: the smallest code at which reconstruction fidelity is essentially saturated and the separability proxy is at its maximum.

1.2 Loss function design

Loss design. v1 used pure MSE. MSE is minimised by the conditional mean, so when fine texture (dune ripples, crater rims, polygonal ground) is uncertain the decoder outputs a blurred average. The v1 reconstructions show this clearly (Fig 1.2). From v2 on, the objective is **0.84·(1 − SSIM) + 0.16·L1 + 0.1·Gradient-L1**, computed only on valid (non-no-data) pixels from v3 on. Its fully differentiable terms (weights in `CONFIGS`) complement each other:

- **(1 − SSIM)** with an 11×11 Gaussian window (σ = 1.5) compares local mean, contrast and structure. It penalises the loss of local structure even when the average brightness is right, which directly targets blur on ridges and crater rims.
- **MSE / L1** anchors absolute brightness. SSIM is insensitive to uniform intensity offsets, so on its own it can drift in albedo.
- **Gradient L1** (finite-difference edges) penalises smeared or misplaced high-frequency edges, which helps sharp linear features such as ridgelines and scarps.

No perceptual loss is used, because every perceptual network we could use would be pretrained, and that is banned.

What the iterations taught us about this loss. Switching from MSE (v1) to this combination (v2) sharpened crater rims and scarps, but it did **not** recover dune-ripple texture: held-out SSIM only went from 0.705 to 0.707. The binding constraint is the bottleneck, not the loss. 256 floats for 51,529 pixels cannot store the exact phase of a ripple field, and a loss cannot recreate information the code has discarded. We keep the structural loss because it spends the unavoidable error on texture detail rather than on structure. Its edge sensitivity, however, made the network lock onto the razor-sharp edges of the no-data wedges (v2's novelty correlated 0.66 with the no-data share). That is why, from v3 on, every term is computed on valid pixels only (Section 0.2).

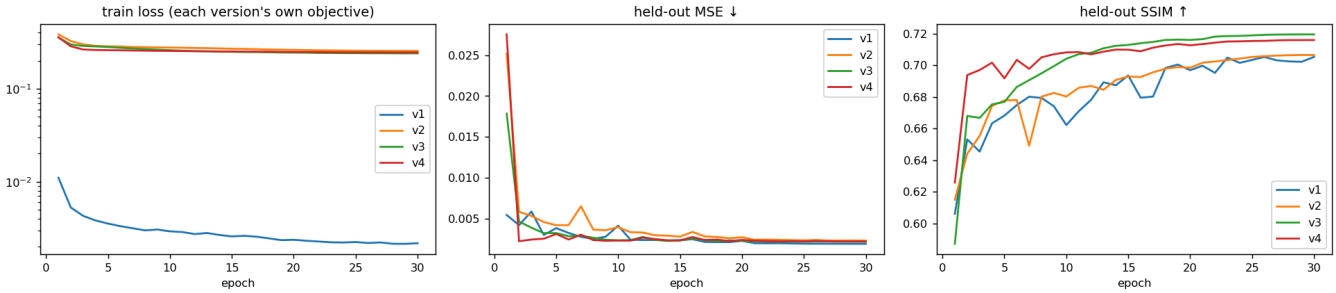


Fig 1.2a - Training curves of v1-v4

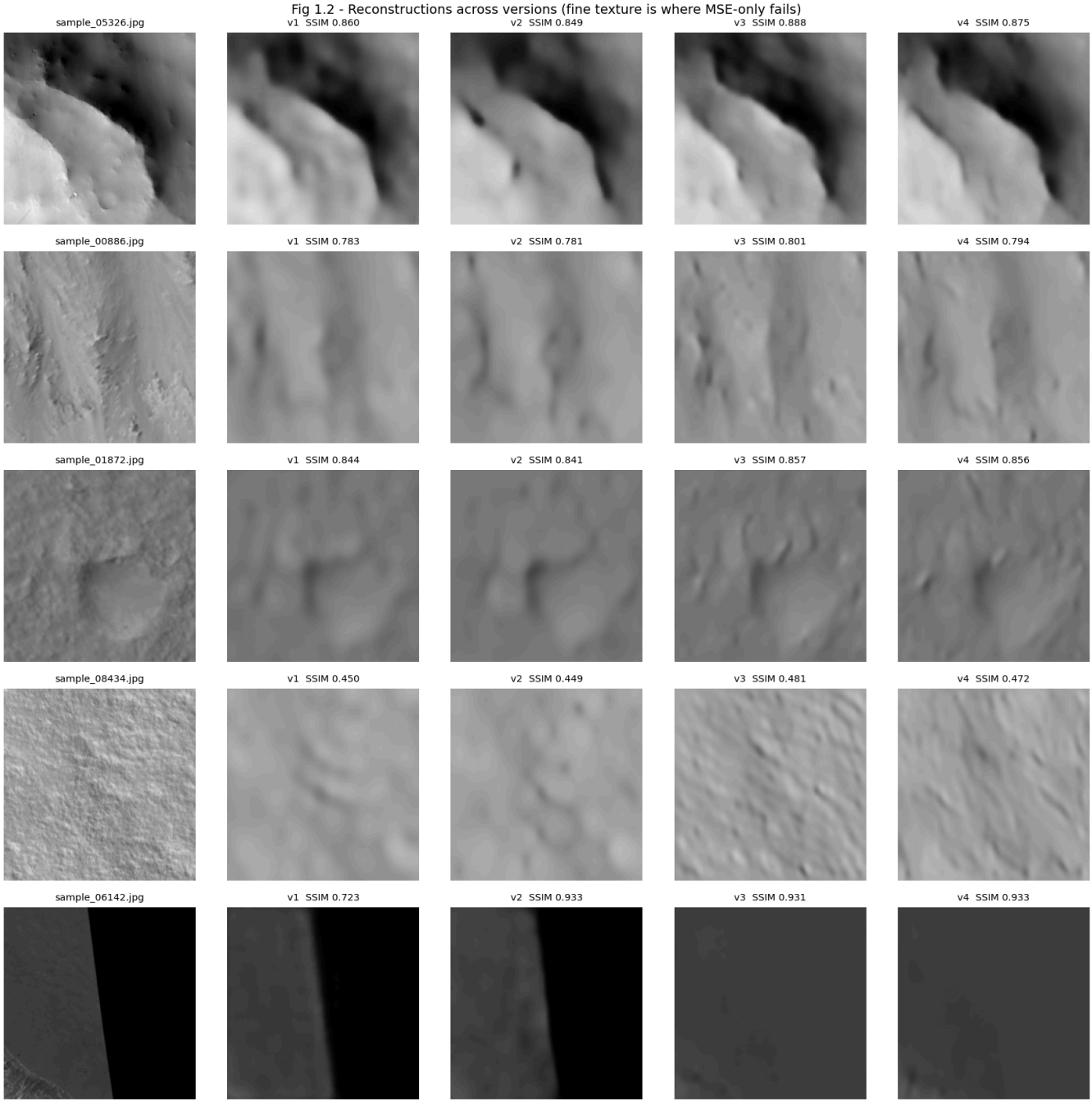


Fig 1.2b - Reconstructions across versions

1.3 Latent space visualisation

PCA (50 components on standardised latents) followed by t-SNE (perplexity 40). Colouring by simple image properties the model never saw as labels shows *which* factors of variation the latent space organises. This is a qualitative diagnostic only, not proof of geological clusters.

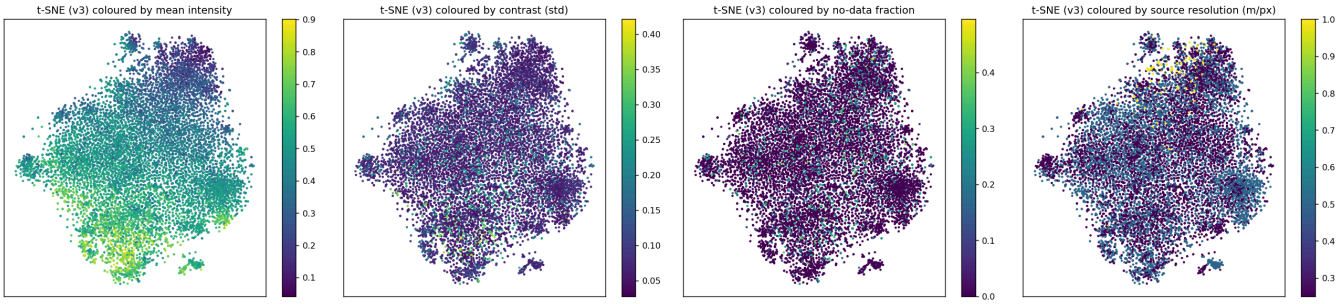


Fig 1.3a - t-SNE of final latents coloured by image properties

Fig 1.3b - t-SNE of v3 latents with nearest-crop thumbnails

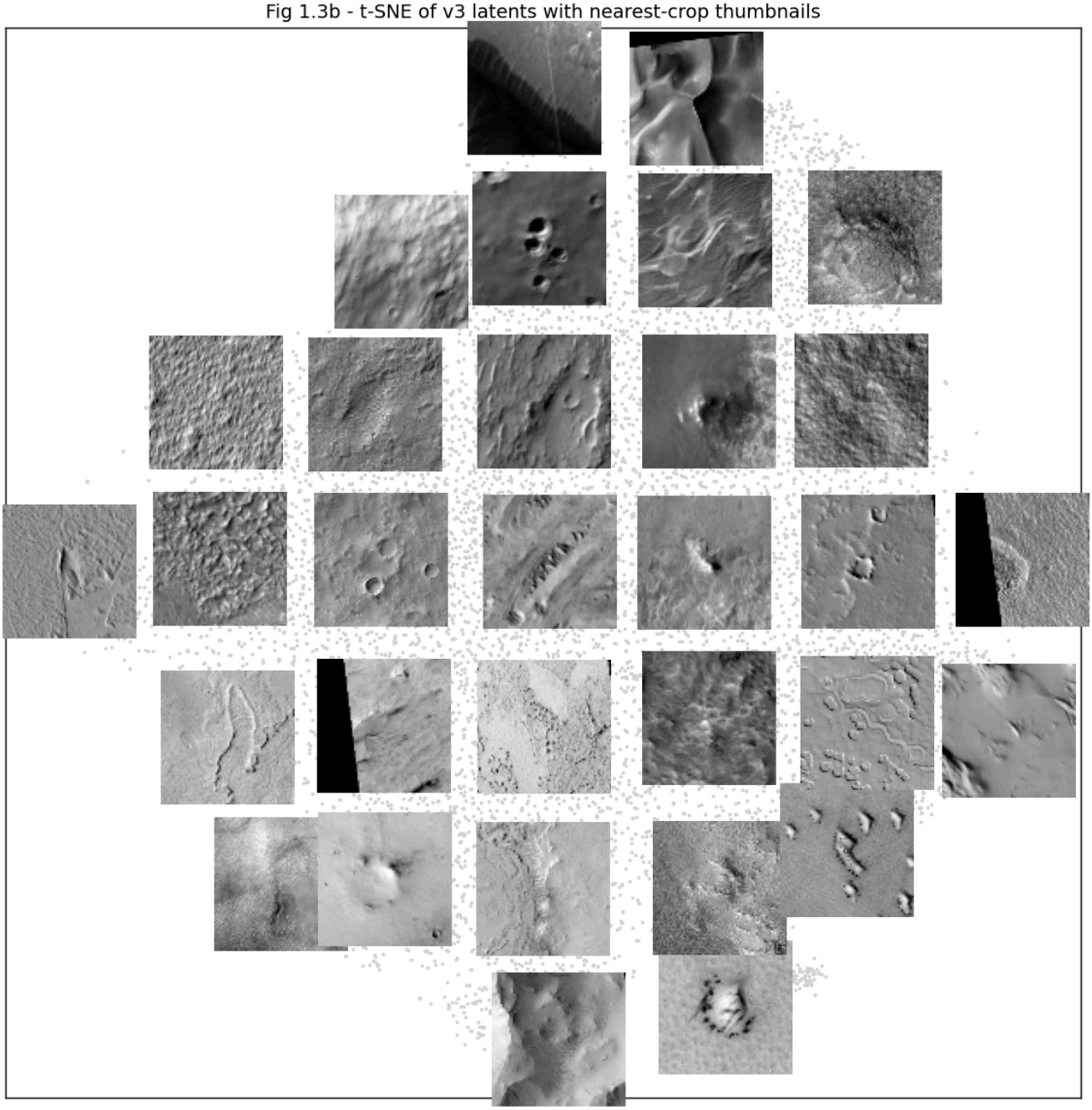


Fig 1.3b - t-SNE with nearest-crop thumbnails

Reading the map (visual interpretation, not labels).

- **No collapse.** The projection is not one structureless blob. It has a dense core, extended arms and several small isolated islands.

- **Brightness is the dominant organising axis.** Mean intensity changes smoothly from bright (lower left) to dark (upper right). Contrast varies more locally.
- **Pixel scale is encoded too.** Crops from the single 1 m/px observation concentrate in one region at the top right, and 0.5 m/px crops are over-represented on the right-hand side. The same terrain imaged at a coarser scale has a different texture frequency.
- **The no-data fraction is spread uniformly.** This is visual confirmation that v3's masking removed the border nuisance that dominated v1 and v2.
- The thumbnails show coherent neighbourhoods rather than discrete clusters: knobby and rough terrain on the left, cratered plains in the centre, ripples and dunes, bright etched or polygonal ground, and smooth dusty slopes. It is a continuum of terrain types.
- After thresholding (Fig 1.3c, produced in Section 2.2), high Novelty Scores lie on the **periphery and in the small islands**, and the dense core scores low, as expected if the Isolation Forest isolates sparse regions. Flagged crops concentrate at the bright lower edge and the dark upper-right edge, which is the visible trace of v3's residual brightness bias (changelog, v4 entry).

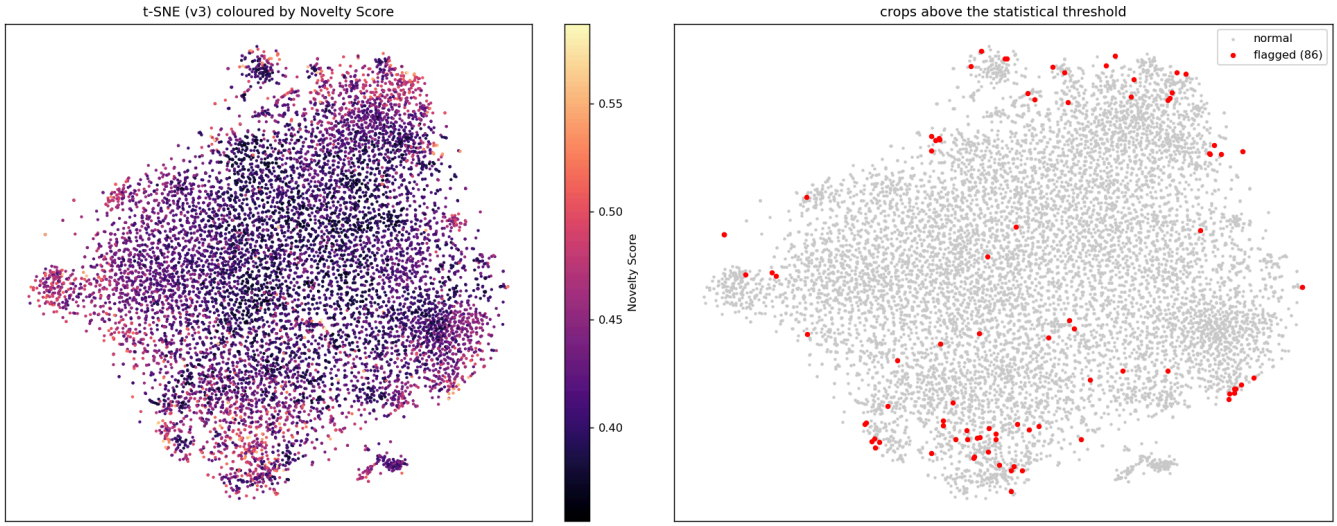


Fig 1.3c - Flagged crops on the t-SNE map

Phase 2 - Isolation Forest Novelty Engine

2.1 Novelty scoring

The Isolation Forest is fitted on the 1-D latent vectors of **all** crops (unsupervised: there is no "clean" training set). We use 500 trees and 256-sample sub-sampling (the original paper's recommended ψ), and average 5 forests with different seeds to reduce the variance of the ranking.

Score convention. sklearn's `score_samples` returns the *negated* anomaly score (higher = more normal). We negate it, so our **Novelty Score = s(x, n) ∈ (0, 1] from Liu et al. (2008), and higher = more anomalous**. This is the competition convention, and it is verified below by the sign of the correlation with sklearn's `decision_function`.

2.2 Statistical thresholding

We want the boundary to come from the **shape of the observed score distribution**, not from an assumed count. First we test whether a Gaussian μ + kσ rule is even appropriate (skewness, kurtosis, D'Agostino K^2 test, Q-Q plot). Then we compare three distribution-driven rules:

1. **Robust modified z-score** (Iglewicz & Hoaglin 1993): M = 0.6745 (s − median) / MAD, flag M > 3.5. Median and MAD have a 50 % breakdown point, so the outliers being hunted cannot inflate the threshold. Under normality 0.6745 × MAD estimates σ, so 3.5 is comparable to ≈3.5σ.
2. **Kneedle elbow** of the ascending sorted-score curve: min-max normalise both axes and take the point of maximum vertical distance below the chord joining the end points. This is the point of maximum curvature, where the curve turns from "bulk" to "tail".
3. **Two-component Gaussian mixture**: the score where the posterior of the high-mean component exceeds 0.5.

mean	std	median	mad	skew	excess_kurtosis	normality_p
0.4229	0.0345	0.4165	0.0217	0.9619	0.9827	0.0000

method	threshold	n_flagged
Modified z-score (median + 3.5 MAD/0.6745)	0.5293	86
Kneedle elbow of sorted scores	0.4597	1513
2-component GMM posterior = 0.5	0.4374	2986
mu + 3 sigma (assumes Gaussian - rejected)	0.5264	95

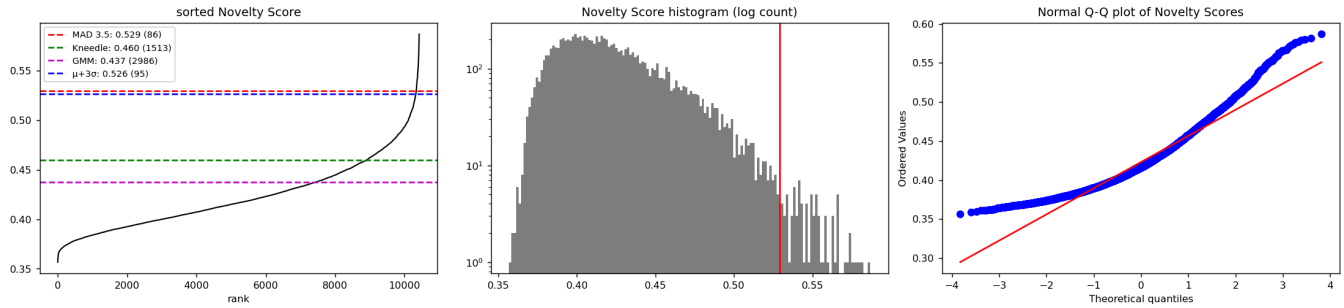


Fig 2.2 - Sorted scores with candidate thresholds, histogram, Q-Q plot

Choice: robust modified z-score > 3.5 → threshold 0.5293, 86 crops flagged (0.83 %)

1. **A Gaussian μ + kσ rule is not supported by the data.** The v3 scores have skewness 0.96, excess kurtosis 0.98 and D'Agostino K^2 = 1,363 (p ≈ 0), and the Q-Q plot bends upward in the right tail. Worse, μ and σ are

estimated *including* the tail we are trying to find, so the outliers inflate their own threshold. We keep μ + 3σ only as a cross-check.

2. **The elbow and GMM boundaries sit inside the body of the distribution, not at a break.** After v3 removed the no-data mode, the score distribution is **unimodal**. A two-component GMM therefore splits the single skewed bulk into two overlapping Gaussians, and its boundary falls near the 71st percentile (2,986 crops). The Kneedle elbow of the *whole* sorted curve marks where the curve leaves its straight middle section, at the start of the ordinary right skew of normal terrain (1,513 crops). Neither boundary coincides with any change in the behaviour of the density.
3. **The modified z-score is built for exactly this situation.** Median and MAD have a 50 % breakdown point, so however many outliers exist they cannot move the threshold. k = 3.5 is the value recommended by Iglewicz & Hoaglin (1993). If the bulk were Gaussian with this robust scale, only about 2 of 10,422 normal crops would exceed it by chance (printed below).
4. **Independent rules agree.** The robust rule (86) and the classical μ + 3σ cross-check (95) land within 10 % of each other. On the sorted-score plot both sit at the foot of the final steep rise (the last ~100 points), where the curve turns from gradual to near-vertical. This is exactly the behaviour an elbow method should key on, once the elbow is sought in the tail rather than across the whole curve.
5. **Sensitivity is reported, not tuned.** The printout shows how the count moves for k = 3 … 5. We keep the conventional k = 3.5 rather than choosing k to hit a count.

> **Final threshold = 0.5293 → 86 crops flagged**

Contamination parameter

sklearn's `IsolationForest(contamination=...)` only changes the offset used by `predict()` to produce ±1 labels. It has **no effect on** `score_samples()`, which is all we use. We set `contamination="auto"` (sklearn's default, offset −0.5), and we never call `predict()`. The final anomaly count is determined solely by the statistical threshold on the continuous Novelty Score above, **independently of any assumed contamination fraction**.

2.3 Image location analysis

Each scored crop is joined to `source_image_metadata.csv` through its `source_image_id` (from `crop_metadata_index.csv`), many-to-one and validated. We then ask whether flags cluster in particular observations, latitudes, illumination conditions, seasons or resolutions.

Metadata resolution. The provided coordinates are quantised. Latitude is on a 5° grid from −45° to 60°, plus the two poles (±90°), and longitude takes only two values, 0° or 180°E (29 and 143 sources). A longitude-latitude map would collapse to two vertical lines, so we analyse **latitude bands × the two longitudes** instead of point locations. We test per-source enrichment with a one-sided binomial test against the global flag rate (Bonferroni-corrected), and test trends with rank statistics (Spearman, Kruskal-Wallis), which make no Gaussian assumption.

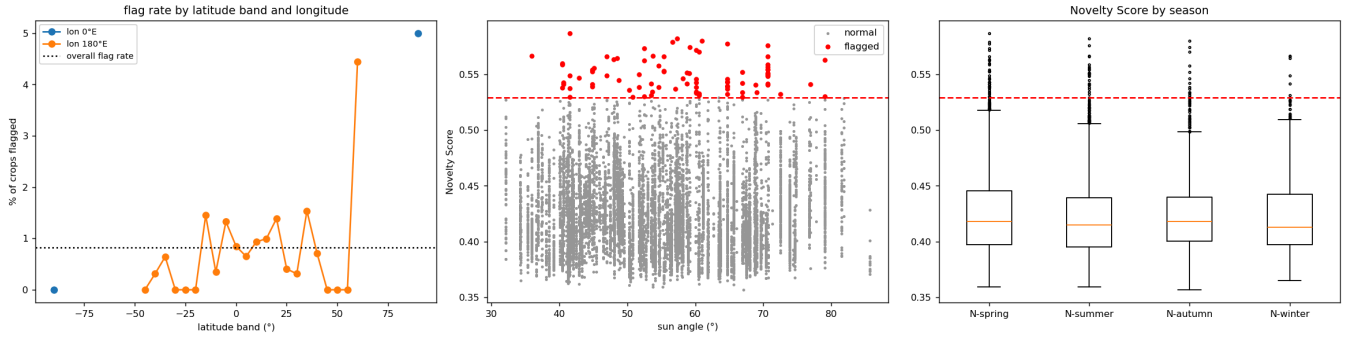


Fig 2.3 - Location, illumination and season

test	value
rho_sun	rho=-0.038, p=0.00012
rho_abslat	rho=0.032, p=0.0012
p_season	0.001644
p_resolution	1.055e-11
p_chi2_season	0.4758
n_src_flagged	39
n_src_enriched	4

source_image_id	n	n_flag	rate	latitude	longitude	sun_angle	season	resolution	p_bonf
SRC_039	16	13	0.8125	90.0000	0.0000	70.6328	N-summer	0.5000	0.0000
SRC_067	92	7	0.0761	-15.0000	180.0000	64.7242	N-spring	0.5000	0.0021
SRC_166	16	6	0.3750	90.0000	0.0000	60.0942	N-spring	0.5000	0.0000
SRC_022	74	4	0.0541	20.0000	180.0000	41.5136	N-spring	0.2500	0.5798
SRC_094	63	4	0.0635	20.0000	180.0000	66.9345	N-autumn	0.2500	0.3224
SRC_140	62	4	0.0645	35.0000	180.0000	44.8556	N-spring	0.2500	0.3039
SRC_131	67	3	0.0448	10.0000	180.0000	60.5405	N-autumn	0.5000	1.0000
SRC_078	28	3	0.1071	15.0000	180.0000	40.4370	N-spring	0.2500	0.2713
SRC_032	14	3	0.2143	-5.0000	180.0000	52.4902	N-spring	0.2500	0.0329
SRC_083	28	3	0.1071	0.0000	180.0000	55.3789	N-spring	0.2500	0.2713

Findings.

- Flags are spread over 39 of the 172 source observations, but **4 sources are significantly enriched** (one-sided binomial test against the 0.83 % base rate, Bonferroni-corrected $p < 0.05$). The two strongest are both **north-polar (90° N)**: SRC_039 has **13 of its 16 crops** flagged and SRC_166 has 6 of 16. Together they contribute 19 of the 86 flags. The other two are SRC_067 (7 of 92, 15° S) and SRC_032 (3 of 14, 5° S).
- **Physical reading: rare terrain, not tampering.** North-polar crops show bright ice, finely banded layered deposits and scarps. For example, the "stripes" of 08731 and 03423 are polar layering, not sensor banding. This terrain is a small minority of the training crops, so its latents are easy to isolate. A mission pipeline would *route* these flags to the polar science team rather than discard them. The metadata is what tells us these flags are a terrain class, not artefacts.
- **Latitude.** Between 45° S and 55° N the flag rate fluctuates around the base rate (0-1.5 %), then jumps at 60° N (≈4.4 %) and 90° N (≈5 %). Correlation with |latitude| overall is negligible ($\rho = 0.03$), so the effect is confined to the high northern bands.
- **Illumination.** ρ(sun angle, novelty) = −0.04. It is statistically significant only because n = 10,422; the effect size is negligible, and flagged crops span the whole 36-79° range.
- **Season.** The score *distributions* differ slightly (Kruskal-Wallis $p = 0.002$), but *flagging* is independent of season (χ^2 $p = 0.48$).
- **Resolution.** Scores differ strongly by resolution ($p \approx 10^{-11}$), consistent with the t-SNE observation that pixel scale changes texture frequency. Yet the lone 1 m/px observation (SRC_111) has only 2 of 137 crops flagged, so scale shifts the bulk of the distribution but does not by itself produce flags.

2.4 Metadata fusion (optional)

We concatenate the standardised image latent with an encoded metadata vector: z-scored sun angle, sin/cos of latitude and longitude (to respect circularity), $\log_2$ resolution, and a one-hot season. We re-score with the same Isolation Forest settings and the same modified-z rule, then compare the flagged sets. The image-only pipeline remains the primary result.

n_image	n_fused	overlap	jaccard	rank_corr
86	93	83	0.8646	0.9921

Fusion changes little: 93 crops are flagged against 86 image-only, with 83 in common (Jaccard 0.86) and a rank correlation of 0.992 between the two scorings. Fusion adds 10 crops and drops 3. Two reasons explain the small effect:

1. The Isolation Forest picks split features uniformly at random, and only 10 of the 266 fused dimensions are metadata, so metadata drives about 4 % of splits.
2. Every crop from one observation shares identical metadata. Metadata can therefore isolate an unusual *observation*, but never a single crop within it, whereas the injected anomalies are single crops.

Metadata is therefore most useful as we used it in 2.3, to *explain* flags after the fact (e.g. the polar enrichment), rather than as an input to detection.

Phase 3 - Reconstruction Interpretability

3.1 Pixel-wise error heatmaps

The five highest-scoring crops **within the statistically thresholded set** are passed back through the decoder. The per-pixel squared error (lightly smoothed, σ = 2 px) is shown alone and overlaid on the original. To make the physical reading less subjective, we also compute descriptors of each error map and image (`heatmap_descriptors`). Each descriptor is reported as a percentile against 500 random un-flagged crops:

- error **concentration** (share of error in the top 5 % of pixels)
- **line score** (a single-row or single-column error spike, as a seam would produce)
- **seam step** (the largest mean-intensity jump across a straight line)
- spectral **periodicity** (stripes or banding)
- **saturation** (clipped pixels)

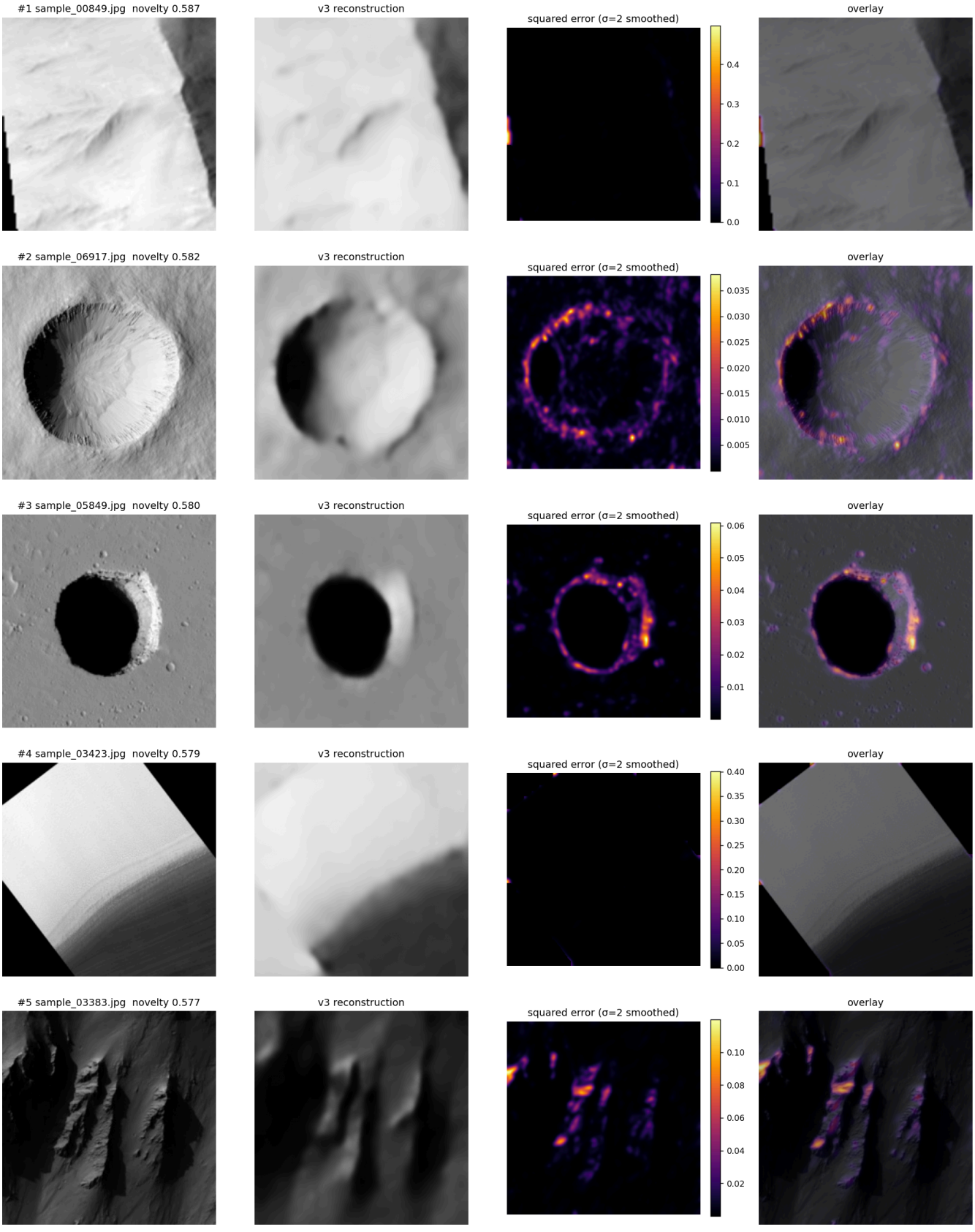


Fig 3.1 - Top-5 thresholded crops: original, reconstruction, error, overlay

rank	filename	source	novelty	recon_mse	concentration	line_score	seam_step	periodicity	saturation
1	sample_00849.jpg	SRC_022	0.5868	0.0023	0.6608	73.9428	0.4180	0.7265	0.0058
2	sample_06917.jpg	SRC_122	0.5819	0.0029	0.2678	2.1512	0.1453	0.7671	0.0033
3	sample_05849.jpg	SRC_008	0.5801	0.0022	0.5052	5.8550	0.1455	0.7792	0.1110
4	sample_03423.jpg	SRC_128	0.5790	0.0010	0.7028	26.7748	0.3874	0.8574	0.0005
5	sample_03383.jpg	SRC_067	0.5773	0.0033	0.5301	4.7143	0.0703	0.4949	0.0195

Percentile of each descriptor against 500 random normal crops:

filename	concentration	line_score	seam_step	periodicity	saturation	mean	std
sample_00849.jpg	99	100	100	89	99	100	100
sample_06917.jpg	33	18	96	92	99	83	100
sample_05849.jpg	93	83	96	93	100	44	100
sample_03423.jpg	99	99	100	98	87	87	100
sample_03383.jpg	94	74	78	66	100	1	98

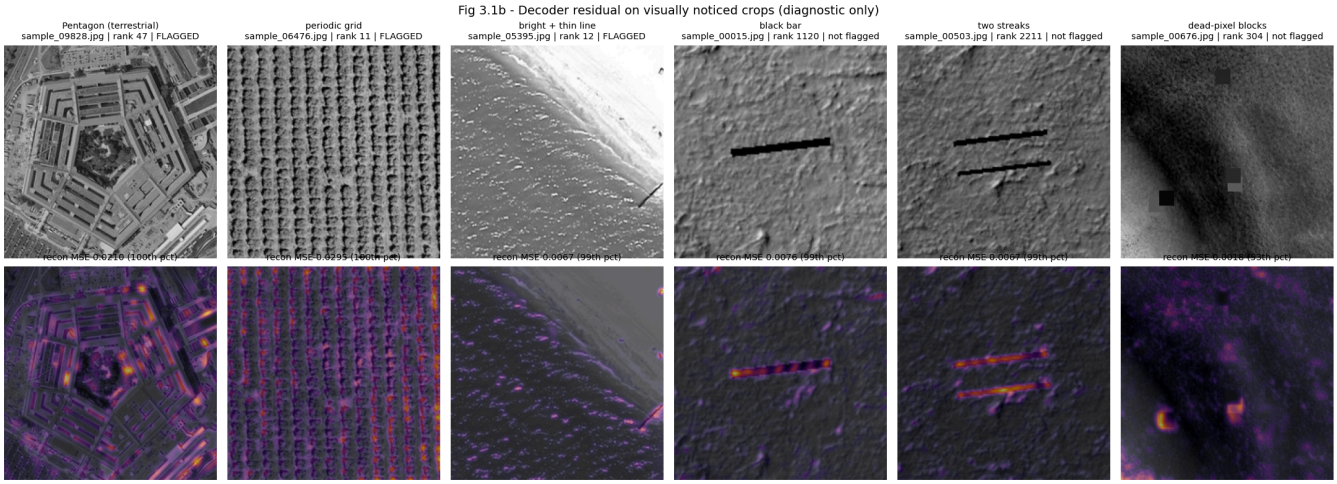


Fig 3.1b - Decoder residual on visually noticed crops (flagged content anomalies vs missed local glitches)

3.2 Geological report

These are **hypotheses**, argued from the heatmap pattern, the descriptors (with percentiles against 500 normal crops) and the metadata. They are not definitive classifications.

#1 sample_00849 (SRC_022, 20° N): straight albedo boundary. Hypothesis: spliced terrain or a mosaic seam. A bright slope with gullies is cut by a *straight*, full-height boundary to a uniformly darker band (seam step 0.42, the 100th percentile; the crop is also brighter than every reference crop). Natural albedo contacts on Mars follow topography and are rarely ruler-straight across a whole frame. A straight contact between two unrelated surfaces is the signature of **semantic splicing**, or, less exotically, of a seam between two HiRISE CCD channels with mismatched gain. The reconstruction error is low (the decoder draws both halves and the boundary), so the flag comes from the *rarity of the configuration* in latent space, not from a failure to encode it. The only concentrated error sits on the edge of the small filled no-data wedge at the left, so part of this crop's score may be the residual no-data effect noted in the changelog.

#2 sample_06917 (SRC_122, 5° S): fresh crater filling the frame. Hypothesis: genuine, rare geology. The error forms a **ring that follows the crater rim**, with streaks down the inner walls. Its concentration is low (33rd percentile), spread along the whole rim. A sharp-rimmed impact crater with gullied walls, a shadowed wall and a bright floor is a high-relief, high-frequency structure that a 256-float global code can only sketch. Such large fresh craters are uncommon at this crop scale, where most crops are plains, dunes and knobby terrain. The pattern is physically coherent (the error sits exactly where relief is steepest) and does *not* resemble any of the injection signatures.

#3 sample_05849 (SRC_008, 25° N): crater or pit with a fully black shadow. Hypothesis: genuine steep-walled depression at the limit of the sensor's dynamic range. 11 % of the pixels are clipped at 0 (saturation at the 100th percentile). The error traces the **rim and the sunlit wall**, where the illuminated rock meets the clipped shadow. The shadow sits on the wall facing away from the sun, consistent with the illumination of the surrounding small craters, which argues for real topography rather than a synthetic black blob. An exposure set for the bright

plain cannot record the shadowed interior, so it clips. This is either a very deep fresh crater or a collapse pit. It is rare, but Martian.

#4 sample_03423 (SRC_128, 90° N): polar layered terrain with a large filled no-data area. Hypothesis: genuine polar geology plus the residual no-data effect. The valid area is a rotated strip, with the two wedges masked and filled. It shows a bright, smooth upper surface and a darker lower surface separated by a curved scarp with fine parallel banding (periodicity at the 98th percentile). At 90° N this is the morphology of the **north polar layered deposits**, stacked ice and dust layers exposed on a scarp. The reconstruction error is the *lowest* of the top 5, so again the novelty is configurational: rare polar terrain, and an unusually large flat fill region, as discussed in the v3/v4 changelog.

#5 sample_03383 (SRC_067, 15° S): dark, low-sun rugged relief. Hypothesis: genuine massifs or ridges under oblique illumination. The frame is darker than 99 % of reference crops, and its contrast is at the 98th percentile. The error is concentrated on the **thin sunlit crests**, not in the shadows. Long shadows from steep spurs make a high-contrast, high-frequency pattern whose exact crest geometry the code cannot store. SRC_067 is one of the four significantly enriched sources (7 flags in 92 crops), which is what we would expect if the whole observation images unusual rugged terrain.

Beyond the top 5 (Fig 3.1b, diagnostic). The heatmap *pattern* separates the injected content from the rare geology above:

- The **Pentagon** aerial photograph (rank 47) and the **periodic grid** (rank 11), which on close inspection resembles an orchard (rows of trees seen from above), have the two highest reconstruction errors in the whole dataset (100th percentile). Their error is **spread diffusely over the entire frame** (concentration 0.18 and 0.16, i.e. almost uniform). The decoder has no Martian vocabulary for rectilinear buildings or a perfect lattice, so it fails everywhere. This is the textbook **non-Martian domain-shift** signature.
- The **black bar** and **two streaks**, which are *not* flagged by the latent score (ranks 1,120 and 2,211), nevertheless sit in the **top 1 % of reconstruction error**, with the error lying *along the straight dark defect* (line scores 4.6 and 5.4). This is the **synthetic sensor-artefact** signature, and it shows the global latent's blind spot for small local defects (changelog, v3 and v4).

heatmap pattern	examples	physical hypothesis
ring following a rim, error on steep walls	06917, 05849	real high-relief feature beyond the code's resolution (rare geology)
error on thin bright crests of a dark frame	03383	oblique illumination of rugged relief
low error, high novelty	00849, 03423	rare *configuration*: straight seam (splice / CCD seam) or polar terrain + large fill
intense error spread over the whole frame	Pentagon 09828, grid 06476	domain shift: non-Martian content
error along a straight dark line	bar 00015, streaks 00503	synthetic sensor artefact (missed by the latent)

Phase 4 - Architecture Iteration & Design Journal

version	val_mse	val_ssim	nodata_mse_ratio	pca90	corr_nov_nodata	n_flag_mad35	flagged_with_nodata
v1	0.0019	0.7053	1.0011	44	0.6347	1596	0.9655
v2	0.0023	0.7065	1.0157	36	0.6571	1985	0.9597
v3	0.0021	0.7195	1.1564	190	0.1100	86	0.4767
v4	0.0022	0.7159	1.1597	139	0.0659	510	0.3353

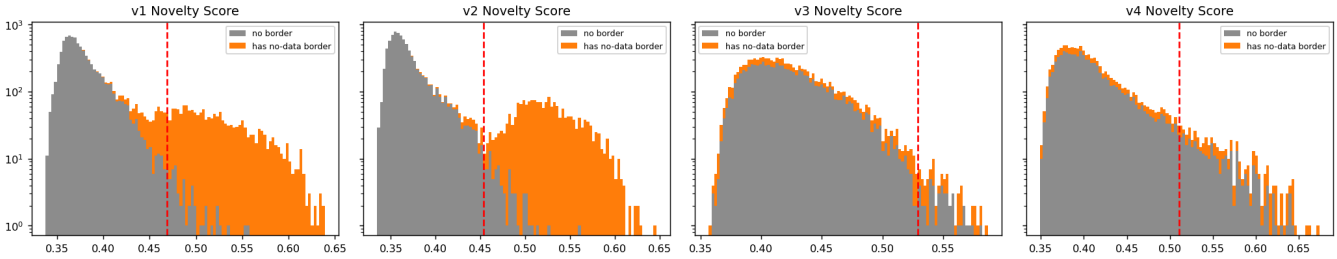


Fig 4.1 - Novelty Score distributions per version (orange = crops with no-data border)

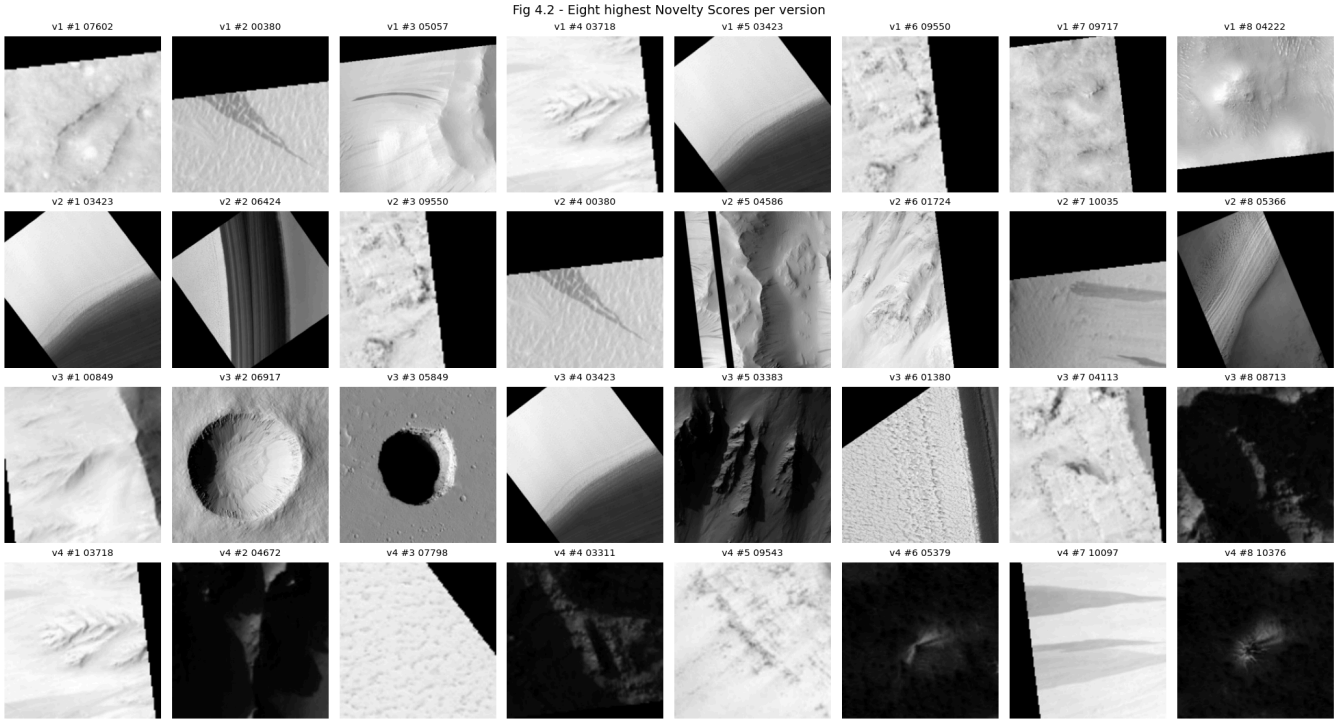


Fig 4.2 - Highest-scoring crops per version

Rank of visually noticed spot-check crops in each version (diagnostic only, never used to set the threshold):

crop	note	v1 rank	v1 flagged	v2 rank	v2 flagged	v3 rank	v3 flagged	v4 rank	v4 flagged
sample_00015.jpg	black bar	4499	False	3859	False	1120	False	2833	False
sample_00503.jpg	two streaks	7419	False	5171	False	2211	False	3158	False
sample_00676.jpg	dead-pixel blocks	1571	True	1915	True	304	False	872	False
sample_09828.jpg	Pentagon (terrestrial)	2875	False	2275	False	47	True	488	True
sample_05508.jpg	terrestrial buildings	2681	False	2494	False	354	False	512	False
sample_06476.jpg	periodic grid	10225	False	5839	False	11	True	178	True
sample_07971.jpg	periodic grid	7163	False	3281	False	21	True	815	False
sample_05395.jpg	bright + thin line	784	True	1735	True	12	True	743	False
sample_00011.jpg	dark terrain (control)	2157	False	1993	False	1297	False	368	True

crop	note	v1 rank	v1 flagged	v2 rank	v2 flagged	v3 rank	v3 flagged	v4 rank	v4 flagged
sample_00385.jpg	dark terrain (control)	4448	False	2992	False	1451	False	1154	False

Every version uses identical data (10,422 crops), a random 90/10 split held out for loss curves, AdamW + one-cycle LR (max 1e-3), batch 64, 30 epochs, horizontal/vertical flip augmentation, bf16 autocast, seed 42, and the same Isolation Forest (500 trees, $\psi = 256$, 5 seeds averaged). Each version changes **one** thing, so its effect can be attributed.

---

v1 - Baseline: convolutional AE, pure MSE

Setup. 5× stride-2 conv encoder (32-64-128-256-256) → Linear → 256-d latent; mirrored transposed-conv decoder; loss = MSE.

Result. Held-out MSE 0.00189, SSIM 0.705 after 30 epochs (plateaued from epoch ~20).

Symptom 1 - Blurry reconstructions, fine texture erased

Dune ripple trains, small-crater fields and rough ground are reconstructed as smooth grey gradients (outputs/v1/recon_grid.png, random crops 07869 and 01503). The error maps of *normal* crops are dominated by texture, not by anything anomalous.

Symptom 2 - The novelty engine ranks map-projection borders, not anomalies

- All of the top-25 Novelty Scores are crops with a black no-data wedge (the edge of the map-projected HiRISE strip). Their error maps are just a bright line tracing the wedge edge.
- Spearman ρ(novelty, reconstruction MSE) = −0.095: what the forest isolates is unrelated to what the decoder fails to explain.
- The score histogram has a long secondary shoulder (0.45-0.63) holding about 15 % of the crops, which is roughly the share of crops with a border. Even the robust modified-z rule (median + 3.5 MAD) therefore flags 1,596 crops (15.3 %), and the elbow and GMM rules flag even more. No defensible threshold can exist while a nuisance factor forms its own mode.
- One latent coordinate has variance 2,800 against a median of 33, so the code spends its largest axis on border geometry.

Diagnosis

1. **Blur:** MSE is minimised by the conditional mean. When the exact phase of a ripple or the rim of a small crater is uncertain, averaging the possibilities gives the lowest expected squared error. The decoder learns to hedge, which erases exactly the high-frequency geology we need in order to notice unfamiliar texture.
2. **Borders:** a no-data wedge is a 0-valued region with a razor-sharp straight edge at an arbitrary angle and position. It is the highest-contrast structure in the dataset. Under MSE it dominates the loss, so the encoder dedicates capacity to encoding wedge geometry. In the latent space, crops with rare wedge angles are then the easiest points to isolate.

Fix

Addressed one at a time: v2 changes only the loss (symptom 1); v3 addresses the no-data nuisance (symptom 2).

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v2 - Structural loss: 0.84·(1 − SSIM) + 0.16·L1 + 0.1·Gradient-L1

Change (loss only; architecture, data and schedule identical to v1).

- **SSIM** (11×11 Gaussian window) penalises lost local contrast and structure even when mean brightness is right, which directly targets blur. The 0.84 / 0.16 split follows Zhao et al. (2017, *Loss Functions for Image Restoration with Neural Networks*).
- **L1** replaces MSE as the intensity anchor. SSIM ignores uniform offsets, and L1 is less dominated by a few huge errors than MSE.
- **Gradient-L1** on finite differences penalises smeared ridgelines and crater rims.

Outcome.

	v1	**v2**
held-out SSIM	0.7053	0.7065
held-out MSE	0.00189	0.00228
PCA components for 90 % latent variance	44	36
Spearman ρ(novelty, no-data fraction)	0.63	**0.66**
crops flagged by modified-z > 3.5	1,596	1,985

Symptom 1 (blur) - essentially unchanged

Crater rims are slightly crisper, but dune ripples (crop 01503) and small-crater fields are still reconstructed as smooth gradients, and held-out SSIM moved by only +0.001.

Diagnosis. The loss was not the binding constraint. The code has only 256 floats for 51,529 pixels (≈200× compression), squeezed through a single linear layer from an 8×8×256 map. The exact phase of a ripple field is high-entropy information that such a code cannot store. Whatever the loss, the decoder can only output the "texture class" with its details averaged out. **Lesson:** the loss decides *how* the unavoidable error is distributed; it cannot create information the bottleneck discarded. For anomaly detection this matters less than it first seemed. The latent only has to encode *what kind of* terrain a crop is, which it can do without reproducing every ripple.

Symptom 2 (no-data borders) - got worse

The novelty histogram (outputs/v2/scores.png) is now clearly **bimodal**. The second mode (0.47-0.60) is almost entirely crops with a black no-data wedge. ρ(novelty, no-data fraction), already 0.63 under v1, rose to 0.66. The number flagged grew from 1,596 to 1,985, and every crop in the top 25 has a wedge.

Diagnosis. Both new loss terms are *edge-sensitive*. The wedge boundary is a 0 → bright step, the strongest gradient in the dataset, so SSIM and gradient-L1 reward the network for placing it exactly. The encoder therefore spends *more* of its (already scarce) capacity on wedge angle and position (the latent spectrum became more concentrated, 44 → 36 PCA components). In latent space, crops with unusual wedge geometry are the easiest to isolate. The wedge is an artefact of map-projecting the HiRISE strip, not terrain, so every one of those flags is a false positive by construction.

Fix → v3

Stop showing the network the wedge. Detect no-data regions, fill them with the crop's median, and exclude them from the loss and error maps, keeping the v2 loss.

---

v3 - No-data masking (data/loss change; architecture and loss weights as v2)

Change. A no-data mask per crop marks near-black (≤ 5) regions that survive a 5×5 opening, touch the crop border and cover at least 0.5 % of it, dilated by 2 px for JPEG ringing. The opening matters: thin interior dark lines, which could themselves be sensor-glitch anomalies, are deliberately **not** masked (outputs/nodata_check.png). Masked pixels are filled with the crop's valid-pixel median and excluded from every loss term, metric and error map. 19.7 % of crops have a mask, covering 18.8 % of those crops on average.

Outcome.

	v2	v3
held-out SSIM (v3: valid pixels)	0.7065	0.7195
Spearman ρ(novelty, no-data fraction)	0.66	**0.11**
Spearman ρ(novelty, reconstruction error)	0.08	**0.44**
PCA components for 90 % latent variance	36	**190**
score distribution	bimodal	unimodal, right-skewed tail
flagged by modified-z > 3.5	1,985 (19 %)	**86 (0.83 %)**
μ + 3σ (cross-check)	45	95

- The nuisance mode is gone. With no second mode, the robust modified-z rule and μ + 3σ now roughly agree (86 vs 95 flagged), which suggests the tail is real rather than an artefact of the rule.
- Novelty now correlates with what the decoder fails to explain (ρ 0.08 → 0.44), and the code uses far more of its dimensions (190 components for 90 % variance instead of 36).
- The flagged set now contains clear content anomalies: an **aerial photograph of the Pentagon** (crop 09828), two **perfectly periodic grid textures** (06476, 07971) and striped strips (08731, 06424) (outputs/v3_flagged_grid.png). Under v1 and v2 these were buried beneath the border crops:

spot-check crop	v1 rank	v2 rank	v3 rank
06476 periodic grid	10,225	5,839	**11**
07971 periodic grid	7,163	3,281	**21**
09828 Pentagon (terrestrial)	2,875	2,275	**47**
05395 bright terrain + thin dark line	784	1,735	**12**

Symptom - small local glitches are ranked far below the threshold

During the sanity pass we noticed crops with obvious synthetic defects: a thick black bar (00015), two black streaks (00503) and square dead-pixel blocks (00676). These are qualitative spot checks we saw by eye; we used them only to diagnose, never to tune the threshold. v3 ranks them 1,120th, 2,211th and 304th, far below the 86 flagged, even though their *reconstruction error* is visibly concentrated on the defect. A secondary residual: 41 of the 86 flagged crops still have a (filled) wedge, against a 20 % base rate. A large perfectly flat region is itself uncommon.

Diagnosis

The bottleneck is one **global** linear layer from the whole 8×8×256 feature map to 256 numbers, and that layer is trained to summarise normal terrain. A defect covering about 2 % of the pixels barely moves this global summary: the encoder projects the crop onto the normal-terrain manifold and "forgets" the bar. The decoder then reproduces clean terrain, which is why the defect shows up in the residual but not in the latent. Latent-density detection with a global code is sensitive to *scene-level* novelty (Earth imagery, alien textures) and blind to *local* defects (glitches, splice seams).

Fix → v4

Keep the information local. Replace the global linear bottleneck with a **spatial bottleneck**, a 1×1 convolution to an 8×8 grid of 4 channels, flattened to the same 256-d 1-D vector. Each coordinate then describes one ~63 px neighbourhood, so a local defect drives a few coordinates to extreme values, which is exactly what an Isolation Forest's axis-aligned random splits isolate fastest.

---

v4 - Spatial bottleneck (architecture change; loss and masking as v3) - REJECTED

Change. The encoder's 8×8×256 map is reduced by a 1×1 convolution to 8×8×4 and flattened to a 256-d 1-D latent. The latent size, compression ratio, loss, masking and schedule are all as in v3. Dropping the two 16,384×256 linear layers cuts the model from 11.1 M to 2.7 M parameters.

Outcome - the hypothesis failed.

	v3	v4
held-out SSIM (valid pixels)	0.7195	0.7159
flagged by modified-z > 3.5	86	510
Spearman ρ(novelty, reconstruction error)	0.44	0.22
Spearman ρ(novelty, deviation from typical brightness)	0.36	**0.66**
share of flagged crops among the 5 % darkest + 5 % brightest	58 %	**97 %**
rank of black bar / two streaks / dead-pixel blocks	1,120 / 2,211 / 304	2,833 / 3,158 / 872
rank of Pentagon / periodic grids	47 / 11, 21	488 / 178, 815

Symptom

The local glitches did **not** move up; they moved *down*. The content anomalies that v3 caught (the Pentagon, the periodic grids) were pushed out of the flagged set, and the flagged set grew six-fold. It is now almost entirely very dark or very bright scenes (outputs/v4_flagged_grid.png), which is exactly the "brightness heuristic" failure the organisers warned about.

Diagnosis

With only 4 channels per cell and no mixing across cells, each coordinate becomes a *local albedo / local gradient* descriptor. An Isolation Forest isolates points that are extreme along individual coordinates, and in a code made of local-albedo features the most extreme crops are simply the darkest and brightest scenes. The injected defects are anomalous in their **configuration** (a perfectly straight dark bar, square blocks, a regular grid), not in their local brightness. At the scale of a 63 px cell, a black bar is about as dark as an ordinary shadow, so none of its coordinates is extreme. The global code of v3 can represent "this scene contains an unnatural structure", because its linear layer combines evidence across the whole crop. The per-cell code cannot.

Decision

Revert to v3 as the final model. v4 stays in the journal as a documented negative result. The choice is made on label-free evidence: v3's novelty agrees better with what the decoder fails to explain (ρ 0.44 vs 0.22), it depends much less on raw brightness, and its score distribution gives a small tail on which the robust and Gaussian rules agree (86 vs 95). The spot checks only corroborate this.

Remaining limitation (honest). Recall is incomplete. A second Earth image (an aerial view of buildings, crop 05508, spotted in v4's grid) ranks 354th under v3 and is not flagged. Small local defects are still missed by *latent-space* scoring in every version. The decoder residual catches two of the three: the black bar and the two streaks are in the **top 1 % of reconstruction error**, with their error lying along the defect (Fig 3.1b). The dead-pixel blocks are invisible even there (53rd percentile). Residual-based or patch-level scoring is therefore a promising extension, but not a complete answer. v3 also keeps some brightness bias: 58 % of its flags lie in the brightness extremes, against a 10 % base rate. Some of that is genuine, because unusually bright or dark terrain (frost, deep shadow) *is* rarer.

Appendix - all flagged crops

filename	source_image_id	novelty	recon_mse	latitude	longitude	sun_angle	season	resolution
sample_00849.jpg	SRC_022	0.5868	0.0022	20.0000	180.0000	41.5136	N-spring	0.2500
sample_06917.jpg	SRC_122	0.5819	0.0029	-5.0000	180.0000	57.3105	N-summer	0.2500
sample_05849.jpg	SRC_008	0.5801	0.0022	25.0000	180.0000	61.0059	N-autumn	0.2500
sample_03423.jpg	SRC_128	0.5789	0.0010	90.0000	0.0000	56.6348	N-spring	0.5000
sample_03383.jpg	SRC_067	0.5773	0.0033	-15.0000	180.0000	64.7242	N-spring	0.5000
sample_01380.jpg	SRC_039	0.5758	0.0043	90.0000	0.0000	70.6328	N-summer	0.5000
sample_04113.jpg	SRC_164	0.5741	0.0027	20.0000	180.0000	59.2269	N-autumn	0.2500
sample_08713.jpg	SRC_032	0.5730	0.0019	-5.0000	180.0000	52.4902	N-spring	0.2500
sample_06963.jpg	SRC_166	0.5716	0.0051	90.0000	0.0000	60.0942	N-spring	0.5000
sample_05326.jpg	SRC_131	0.5700	0.0010	10.0000	180.0000	60.5405	N-autumn	0.5000
sample_06476.jpg	SRC_158	0.5665	0.0295	-5.0000	180.0000	35.9541	N-winter	0.2500
sample_05395.jpg	SRC_024	0.5664	0.0067	10.0000	180.0000	53.7258	N-summer	0.2500
sample_05994.jpg	SRC_072	0.5661	0.0029	15.0000	180.0000	46.9668	N-winter	0.2500
sample_08731.jpg	SRC_039	0.5660	0.0027	90.0000	0.0000	70.6328	N-summer	0.5000
sample_02344.jpg	SRC_083	0.5660	0.0020	0.0000	180.0000	55.3789	N-spring	0.2500
sample_02790.jpg	SRC_101	0.5643	0.0044	5.0000	180.0000	48.4794	N-winter	0.2500
sample_00307.jpg	SRC_047	0.5632	0.0015	0.0000	180.0000	48.0287	N-spring	0.5000
sample_01094.jpg	SRC_111	0.5627	0.0026	-35.0000	180.0000	79.0699	N-spring	1.0000
sample_09453.jpg	SRC_032	0.5625	0.0040	-5.0000	180.0000	52.4902	N-spring	0.2500
sample_07744.jpg	SRC_078	0.5597	0.0020	15.0000	180.0000	40.4370	N-spring	0.2500
sample_07971.jpg	SRC_078	0.5589	0.0157	15.0000	180.0000	40.4370	N-spring	0.2500
sample_04696.jpg	SRC_039	0.5586	0.0042	90.0000	0.0000	70.6328	N-summer	0.5000
sample_02718.jpg	SRC_123	0.5578	0.0054	0.0000	180.0000	54.5981	N-autumn	0.5000
sample_08621.jpg	SRC_039	0.5568	0.0048	90.0000	0.0000	70.6328	N-summer	0.5000
sample_05036.jpg	SRC_104	0.5557	0.0022	25.0000	180.0000	45.1020	N-summer	0.5000
sample_06424.jpg	SRC_039	0.5545	0.0033	90.0000	0.0000	70.6328	N-summer	0.5000
sample_07875.jpg	SRC_140	0.5541	0.0014	35.0000	180.0000	44.8556	N-spring	0.2500
sample_07170.jpg	SRC_039	0.5541	0.0048	90.0000	0.0000	70.6328	N-summer	0.5000
sample_04017.jpg	SRC_083	0.5529	0.0019	0.0000	180.0000	55.3789	N-spring	0.2500
sample_00606.jpg	SRC_083	0.5526	0.0014	0.0000	180.0000	55.3789	N-spring	0.2500
sample_08200.jpg	SRC_140	0.5526	0.0014	35.0000	180.0000	44.8556	N-spring	0.2500
sample_09543.jpg	SRC_094	0.5519	0.0016	20.0000	180.0000	66.9345	N-autumn	0.2500
sample_09777.jpg	SRC_070	0.5516	0.0021	60.0000	180.0000	58.7890	N-summer	0.5000
sample_03118.jpg	SRC_155	0.5510	0.0071	-5.0000	180.0000	59.1175	N-summer	0.2500
sample_03708.jpg	SRC_039	0.5510	0.0040	90.0000	0.0000	70.6328	N-summer	0.5000
sample_07663.jpg	SRC_039	0.5510	0.0043	90.0000	0.0000	70.6328	N-summer	0.5000
sample_02760.jpg	SRC_052	0.5499	0.0050	5.0000	180.0000	51.7054	N-summer	0.2500
sample_05733.jpg	SRC_039	0.5497	0.0040	90.0000	0.0000	70.6328	N-summer	0.5000
sample_03718.jpg	SRC_022	0.5490	0.0016	20.0000	180.0000	41.5136	N-spring	0.2500
sample_09357.jpg	SRC_072	0.5489	0.0014	15.0000	180.0000	46.9668	N-winter	0.2500
sample_09067.jpg	SRC_039	0.5489	0.0041	90.0000	0.0000	70.6328	N-summer	0.5000
sample_04629.jpg	SRC_039	0.5482	0.0049	90.0000	0.0000	70.6328	N-summer	0.5000
sample_08138.jpg	SRC_028	0.5467	0.0012	10.0000	180.0000	42.8948	N-spring	0.2500
sample_06777.jpg	SRC_114	0.5463	0.0017	-10.0000	180.0000	58.1765	N-autumn	0.5000

filename	source_image_id	novelty	recon_mse	latitude	longitude	sun_angle	season	resolution
sample_01653.jpg	SRC_067	0.5460	0.0030	-15.0000	180.0000	64.7242	N-spring	0.5000
sample_07428.jpg	SRC_166	0.5457	0.0033	90.0000	0.0000	60.0942	N-spring	0.5000
sample_09828.jpg	SRC_156	0.5448	0.0210	10.0000	180.0000	48.7374	N-summer	0.2500
sample_09498.jpg	SRC_067	0.5434	0.0033	-15.0000	180.0000	64.7242	N-spring	0.5000
sample_04516.jpg	SRC_067	0.5430	0.0019	-15.0000	180.0000	64.7242	N-spring	0.5000
sample_05582.jpg	SRC_166	0.5426	0.0046	90.0000	0.0000	60.0942	N-spring	0.5000
sample_06820.jpg	SRC_106	0.5424	0.0028	25.0000	180.0000	40.6390	N-spring	0.5000
sample_10319.jpg	SRC_094	0.5420	0.0015	20.0000	180.0000	66.9345	N-autumn	0.2500
sample_08392.jpg	SRC_070	0.5416	0.0022	60.0000	180.0000	58.7890	N-summer	0.5000
sample_06428.jpg	SRC_039	0.5416	0.0055	90.0000	0.0000	70.6328	N-summer	0.5000
sample_08001.jpg	SRC_139	0.5414	0.0035	40.0000	180.0000	53.5333	N-winter	0.2500
sample_07701.jpg	SRC_136	0.5412	0.0170	-35.0000	180.0000	76.9224	N-spring	0.5000
sample_06351.jpg	SRC_106	0.5412	0.0021	25.0000	180.0000	40.6390	N-spring	0.5000
sample_01724.jpg	SRC_140	0.5410	0.0021	35.0000	180.0000	44.8556	N-spring	0.2500
sample_03918.jpg	SRC_108	0.5407	0.0025	-40.0000	180.0000	68.8563	N-summer	0.5000
sample_05584.jpg	SRC_039	0.5407	0.0049	90.0000	0.0000	70.6328	N-summer	0.5000
sample_09975.jpg	SRC_067	0.5393	0.0009	-15.0000	180.0000	64.7242	N-spring	0.5000
sample_00473.jpg	SRC_067	0.5390	0.0016	-15.0000	180.0000	64.7242	N-spring	0.5000
sample_05368.jpg	SRC_140	0.5389	0.0015	35.0000	180.0000	44.8556	N-spring	0.2500
sample_06335.jpg	SRC_052	0.5386	0.0010	5.0000	180.0000	51.7054	N-summer	0.2500
sample_04902.jpg	SRC_123	0.5386	0.0060	0.0000	180.0000	54.5981	N-autumn	0.5000
sample_05284.jpg	SRC_166	0.5385	0.0034	90.0000	0.0000	60.0942	N-spring	0.5000
sample_01216.jpg	SRC_078	0.5382	0.0027	15.0000	180.0000	40.4370	N-spring	0.2500
sample_10097.jpg	SRC_022	0.5374	0.0008	20.0000	180.0000	41.5136	N-spring	0.2500
sample_06196.jpg	SRC_103	0.5372	0.0037	-35.0000	180.0000	57.0904	N-autumn	0.2500
sample_07084.jpg	SRC_168	0.5367	0.0017	90.0000	0.0000	64.7096	N-summer	0.2500
sample_06410.jpg	SRC_074	0.5358	0.0029	0.0000	180.0000	50.2546	N-spring	0.5000
sample_00413.jpg	SRC_050	0.5342	0.0017	0.0000	180.0000	53.6801	N-summer	0.2500
sample_06220.jpg	SRC_166	0.5341	0.0015	90.0000	0.0000	60.0942	N-spring	0.5000
sample_07299.jpg	SRC_094	0.5341	0.0030	20.0000	180.0000	66.9345	N-autumn	0.2500
sample_03147.jpg	SRC_058	0.5339	0.0009	30.0000	180.0000	69.0609	N-autumn	0.5000
sample_10203.jpg	SRC_166	0.5332	0.0032	90.0000	0.0000	60.0942	N-spring	0.5000
sample_03515.jpg	SRC_131	0.5330	0.0011	10.0000	180.0000	60.5405	N-autumn	0.5000
sample_05004.jpg	SRC_091	0.5324	0.0031	-40.0000	180.0000	72.5113	N-summer	0.5000
sample_00824.jpg	SRC_067	0.5323	0.0012	-15.0000	180.0000	64.7242	N-spring	0.5000
sample_08152.jpg	SRC_131	0.5312	0.0004	10.0000	180.0000	60.5405	N-autumn	0.5000
sample_01916.jpg	SRC_139	0.5311	0.0034	40.0000	180.0000	53.5333	N-winter	0.2500
sample_06380.jpg	SRC_111	0.5304	0.0040	-35.0000	180.0000	79.0699	N-spring	1.0000
sample_04060.jpg	SRC_032	0.5303	0.0010	-5.0000	180.0000	52.4902	N-spring	0.2500
sample_05800.jpg	SRC_023	0.5297	0.0085	-10.0000	180.0000	50.8374	N-spring	0.2500
sample_10092.jpg	SRC_022	0.5295	0.0009	20.0000	180.0000	41.5136	N-spring	0.2500
sample_02823.jpg	SRC_094	0.5295	0.0008	20.0000	180.0000	66.9345	N-autumn	0.2500